



Don Bosco Institute of Technology, Mumbai

Department of Computer Science

Project Title:

STOCKVISION

Analyze · Visualize · Interpret | Market Trend Insights using Smart Money Concepts

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ABSTRACT

StockVision is a Python-based stock market analysis and visualization system designed to study price movements using graphical tools such as candlestick and line charts. It incorporates Smart Money Concepts like liquidity, supply-demand zones, and price imbalance to identify potential trading opportunities. The system focuses on trend-based insights rather than exact prediction, helping users understand market behavior effectively.



BACKGROUND



- Rapid growth in stock market participation
- Lack of structured analysis tools for beginners
- Problem:** Traders fail due to poor understanding of market trends and psychology

INPUT PARAMETERS

Stock Symbol (e.g., NIFTY, RELIANCE)
Timeframe (5 min / Daily)
Historical Price Data
Volume & Market Structure Data
Demo Trading Balance (Paper Money)

OBJECTIVES

- Analyze market trends using data visualization
- Identify high-probability trade zones
- Provide a risk-free paper trading environment

METHODOLOGY

- Data Collection (historical stock data)
- Data Processing using Python
- Visualization using libraries (Matplotlib / Plotly)
- Market Trend Analysis
- Smart Money Concept Detection
- Paper Trade Simulation

CORE CONCEPTS

Imbalance of Power (Candle Psychology)

Supply & Demand Zones

Price Imbalance (FVG & Breakaway Gap)

Liquidity (Grab & Sweep)

Stop Loss Hunt

LIVE OUTPUT (DEMO)

Trend: **Bullish**

Entry Zone: ₹2450-₹2465

Stop Loss: ₹2420

Target: ₹2550

Mode: Demo Trading (Paper Money)

P&L: +₹3,200 (Virtual)

Badge: **High Probability Setup**

LIVE MARKET DASHBOARD



RESULTS

- Clear visualization of price movements
- Better understanding of market structure
- Improved decision-making through analysis



SUMMARY

- Data-driven market insights
- Combines theory + practical simulation
- Beginner-friendly and effective



FUTURE SCOPE

- AI-based predictive analytics
- Real-time stock market API integration
- Mobile application development
- Advanced portfolio tracking



Powered by:

Python:- Tkinter, Matplotlib, yfinance, pandas, numpy



NOTE: "This project uses simulated (paper) trading. No real money is involved."